

ALEX MADARAS

BACK-END DEVELOPER

✉ alexandrumadaras00@gmail.com

☎ (+40)724346769

in alexmadaras

🐙 alexmadaras00



I'm a 24-year-old Computer Science Master's graduate and aspiring Software Engineer with a passion for building meaningful software, particularly mobile and full-stack web applications that enhance people's everyday lives, some of which I've showcased in my [web portfolio](#). I'm driven by curiosity and enjoy experimenting with new technologies, often integrating them into both personal and professional projects. I thrive in collaborative teams where innovation, learning, and shared motivation are valued.

EDUCATION

University of Twente 2022 – March 2025

Master's in Computer Science

Courses: Software Engineering Techniques, Software Management, Requirement Elicitation, Software Architecture, Microservices Architecture in Web Development, Machine Learning, Exploratory Data Analysis.

West University of Timisoara 2019–2022

Bachelor's in Computer Science

Courses: Algorithms and Data Structures, Introduction to Python, Software Engineering, Practice Stage, Statistics and Numerical Analysis, Web Technologies, Artificial Intelligence, Object-Oriented Programming

IADE Creative University 2020–2021

*Erasmus+ Semester in Lisbon, Portugal–
Bachelor of Game and Apps Development*

Courses: Collaborative Learning, Mobile Programming in Kotlin, Game Frameworks, Agile Management for Multiplatform Projects.

Focus: building a Multiplatform Project consisting of a Windows 2D Game in Unity connected to a Companion App in Android through an SQL Database.

SKILLS

Programming: Java, Kotlin, Python, SQL, Node.js

Technical: Android, Firebase, GIT, Machine Learning, Data Analysis, Algorithms and Data Structures

Languages: English (C1 Cambridge), Dutch (NT2 A1), Romanian (Mother tongue)

Soft Skills: Communicative, Problem Solving, Good Listener, Ambitious, Attention to Details, Critical Thinking

Interests: Software/Mobile Development, AI, Tour Guiding and Tourism, History, Calisthenics, Psychology

ACHIEVEMENTS AND LEADERSHIP

- **Hackathon:** 2nd place at IADE Creative Hackathon 2021 – designed & developed a mobile app to support Erasmus students in Lisbon.
- **Tour Guiding:** Certified local tour guide since 2021; volunteered at Timisoara's Cultural Capital events, guiding tourists from 30+ countries.
- **Sports:** Captain of a university 7v7 football and mixed volleyball team.
- **Volunteering:** Participated in Erasmus+ projects on sustainability (YouRoTrip 2022) and alternative learning (GeoFun 2022).

WORK EXPERIENCE

ING Nederland B.V. Apr 2024 – Nov 2024

AI Engineer Intern

- Researched AI algorithms to detect data quality issues in credit risk data for my **Master's thesis** (graded 8.0/10).
- Performed **Exploratory Data Analysis** on ING's **credit risk data set**, while improving my business understanding, communication and delivery skills through presentations, including to CIO-level stakeholders.
- Built, trained, and validated an ensemble of one machine learning model (**Isolation Forest**) and a deep learning model (**Autoencoders**) as a Proof of Concept at ING, now continued by another team.

Victory Square Partners Jul 2021 – Sept 2021

Android Developer Intern

- Developed practical **Android** skills using **MVVM** architecture within an **Agile Scrum** team, applying **GIT** for effective version control and collaboration.
- Contributed to the migration of a mobile door lock app to an MVVM architecture, while enhancing the UI of **50%** of key screens in the Android app, improving usability and user satisfaction.
- Implemented **Firebase Analytics** to track user behavior, enabling data-driven improvements.

REMARKABLE PROJECTS

Trip Palette Feb 2024 – Apr 2024

Team Lead

- Built a full-stack web app using **SOA microservices** with **Java** (back-end), **HTML/Thymeleaf/JavaScript** (front-end), and **PostgreSQL**, focusing on modularity and maintainability.
- Containerised services with **Docker** for efficient, isolated deployment.
- Integrated external APIs, including **OpenAI**, via **GET/POST** requests for seamless third-party interaction.

Java Windows PC Game – AzulApp

Java Engineer Apr 2023 – Jun 2023

- Developed a **Java** desktop game in a team of 4 using **TDD**, **Clean Code** principles, and **MVC architecture**.
- Implemented core gameplay logic, a **GitLab CI/CD** pipeline (YAML), and applied 4 design patterns (Singleton, Observer, Proxy, Facade).
- Achieved **96.89%** test coverage with JUnit and earned a **4.4/5** code quality score on **Sigrid**, indicating a high-quality codebase.